

Response to Reviewers

CLDY-D-25-00309: Lorenz Energy Cycle Climatology for the
Southwestern Atlantic Cyclones

Response to Reviewer 1

Overview:

This manuscript analyses the Lorenz Energy cycle for > 6500 cyclones over the south Atlantic using ERA5 reanalysis data. Compared to previous studies focusing on the Lorenz energy cycle this study uses a Lagrangian limited area approach. The manuscript also attempts to understand the variability in the Lorenz energy cycle and determine how different types of intense cyclones differ in terms of their energy cycle. Overall, the manuscript is well in scope of the journal and is interesting.

However, some parts of the methods are not explained clearly enough making it hard to really understand the analysis. At points I was also missing a clear physical interpretation of some of the results. Lastly, one of the key results, that the barotropic conversion terms acts to transfer energy from the kinetic energy of the zonal flow to the eddy kinetic energy (opposite to what is expected) is surprising and not explained. This needs to be addressed in the manuscript and a reader needs to be convinced that this result is physically meaningful rather than an artifact of the method. Below I list a number of additional points that need to be considered before this manuscript can be considered for publication.

Comments:

1. Abstract. The results that are presented in the abstract are vague, for example, “Our results reveal consistent patterns of baroclinic and barotropic conversions along with diabatic processes, with distinct energetic signatures characterising each life cycle state”. More details regarding what are the distinct differences between the different stages could be added.

R: The abstract has been revised to specify the key energetic differences between the intensification, mature, and decay stages, explicitly quantifying the dominant baroclinic, barotropic, and diabatic contributions for each phase.

2. Introduction, lines 74-75. There are quite a few studies which have examined the Lorenz energy cycle for baroclinic wave simulations. These are relevant to this study and could be cited here. For example, Rantanen et al (2019), Kirshbaum et al (2018).

R: The suggested references have been added (L 84–85).

3. Line 107. “over 7000 cyclones... were analysed”. Check this statement as it seems inconsistent with text later on in the manuscript.

R: The statement has been corrected to "over 6700 cyclones were analysed" to ensure consistency with the rest of the manuscript (L119).

4. Section 2.1, cyclone tracking. Does this database cover all seasons and are all seasons analysed here together? Would it make more sense to consider each season separately?

R: Yes, the database covers all seasons, and we analysed them together. This information was not included in the manuscript to avoid unnecessary details. We agree that considering each season separately would be a highly relevant analysis, but the current manuscript is already quite lengthy. Therefore, we plan to address this in a future study (see de Souza et al., 2024a: the results are already prepared, and we just need to organize them into a new manuscript). We have added this point at the end of the conclusions section (L812-816).

5. Section 2.1 cyclone tracking. Is there a minimum threshold for the 850-hPa relative vorticity for a cyclone to be detected? Usually when TRACK is used, only cyclones with a maximum T42 relative vorticity greater than $1 \times 10^{-5} \text{ s}^{-1}$ are retained.

R: Yes. We retained only cyclones whose maximum T42-filtered 850-hPa relative vorticity exceeds $1 \times 10^{-5} \text{ s}^{-1}$, exactly as in the standard TRACK configuration. Note that vorticity is scaled by -1 in the Southern Hemisphere. All tracking details follow the procedure fully documented in Gramscianinov et al. (2020); we have added this reference in the manuscript to guide the reader (L145-146).

6. Figure 1, are the units correct here? The maximum values are > 100 cyclones per month which is unrealistically high. Possibly these values are per year or per season? Please check.

R: The values are correct, please see de Souza et al. (2024b). In traditional cyclone studies, cyclone density is often shown for a single time step or is normalized by tracks (e.g., Hoskins and Hodges, 2002). In our case, we are showing the density for all time steps within 5° spherical cap. Cyclones may form inside the cap, move into it from adjacent areas, linger over several time steps, or traverse it entirely; each of these time-step occurrences is counted, so the > 100 figure reflects multiple passages rather than distinct individual storms. We understand that the use of "cyclones per month" can cause confusion, so we have changed it to "cyclonic features" in the revised manuscript (caption of Figures 1 and 10).

7. Line 151 “Due to data availability issues in the ERA5 database...” Can this be explained / justified? As far as I am aware there are no access issues with ERA5 at the moment. This appears to be a large issue as it results in 1142 extratropical cyclones being excluded from the analysis which corresponds to almost 15% of all cyclones.

R: During the second half of 2023 and the first half of 2024, ECMWF was migrating its data infrastructure and updating its API services. Throughout this period, the ERA5 API was frequently unstable, with slow data transfers and repeated interruptions caused by server outages or other unexplained errors. These technical problems often resulted in incomplete or failed downloads, despite multiple attempts and automated retries. As a result, some cyclones

could not be included in the analysis. This limitation was therefore due to temporary external technical issues with the data access infrastructure, not with the underlying data or methods. We have clarified this in the revised manuscript (L163-174)

8. Section 2.2. Additional mathematical details are needed here concerning how the terms are computed, in particular how the zonal / areas means are computed. This information could be in an appendix. I also found the text in lines 162 – 174 very hard to read. It may help a reader to include a table where all of these terms are defined.

R: We have revised this paragraph for clarity (L179-196) and added an appendix that now contains all governing equations, including the zonal/area-mean operators.

9. Section 2.2. Based on the text I cannot understand what the boundary pressure work terms are and if they are physically meaningful or just a mathematical aspect to these equations.

R: The boundary-pressure-work terms are genuine physical fluxes (mechanical work done by winds against pressure at the domain edges); they are not a mathematical artefact. To our knowledge, no study has yet provided a detailed dynamical interpretation of these terms (hence we do not extend that discussion here), but we now (i) clarify their definition and physical meaning in the revised paragraph of Sect. 2.2 (L193-196), and (ii) comment on their possible role in the energy balance in the last paragraph of Sect. 4.1 (L663-670).

10. Section 2.2. It is unclear how these equations deal with a domain that is moving. Can further explanation be added to explain which terms account for this (rather than just the fact that the domain is local not global).

R: In Michaelides' semi-Lagrangian formulation, the fields are evaluated in a frame that moves with the cyclone, but the governing balances themselves are identical to the fixed-Eulerian case; i.e. no extra "motion" terms appear and any changes introduced by the shifting window are implicitly contained in the boundary-flux terms. We have added a discussion to clarify this point and to note its limitations (L760–772).

11. Line 185 "Semi-Lagrangian" why is this framework "semi-Lagrangian" and not just Lagrangian?

R: A purely Lagrangian method would track the movement of all air parcels (see Federer et al., 2024), whereas in our case, only the spatial domain is moving. This distinction is not explained in the manuscript, as it is already discussed in the original article we cite when mentioning the Semi-Lagrangian method (Michaelides, 1999). Additionally, we believe that this differentiation is not crucial for the scope of our current study.

12. Section 2.3, lines 210-214. TRACK usually smoothes the vorticity field when the cyclone tracks are identified – this is done by truncating the input relative vorticity to T42 spectral resolution to ensure that small scale features are removed and only synoptic-scale ETCs are identified. Therefore, it is unclear to me why these additional filters (Lancoz and Savitzky- Golay filters) are needed. If these are needed, additional details about these two filters should be included.

R: You are correct, and there is no need for the additional preprocessing steps. The passage was only explaining how the program works, but the Lanczos and Savitzky-Golay filters are optional and were not used in our analysis, as described in the article we cited, where the cyclone phase detection method is detailed. To avoid any confusion, we have removed this passage from the manuscript (L241-248).

13. Line 227. It is unclear what variables are used as input to the EOF /PCA analysis and also whether all times of all cyclone lifecycles are used as input.

R: The EOF/PCA was performed on the complete set of Lorenz-energy-cycle terms. For every cyclone, we first averaged each term over its four objectively detected life-cycle phases; the resulting phase-mean values constituted the input matrix for the EOF calculation. All cyclones and all phases were therefore included. We have clarified this in Sect. 2.3 (L264-266).

14. Line 244. There is a small typo here. I think it should be “from identified systems where at Least...”

R: We revised this sentence to make it clearer (L274-276).

15. Lines 253 – 265. K-means. I am unsure of the motivation for this part of the analysis. The stated goal is “For determining the LEC patterns of the most intense systems”, however I think this could be done without the k-means clustering. The method selects the systems exceeding the 99th percentile of maximum vorticity and therefore by looking at these alone the question stated could be answered.

R: Our aim was not only to single out the strongest cyclones but also to test whether intense systems fall into energetically distinct subsets (e.g. predominantly baroclinic vs. barotropic, or mixed). When K-means was applied to the entire population, it mainly reproduced clusters that differed only in overall intensity. By first restricting the sample to the upper 10% in peak ζ_{850} (**see answer for question 25**) we removed that bias; the clustering algorithm could then separate the intense cyclones according to their internal LEC structure (as expressed in Fig. 12). In this way we showed that even among intense storms the relative roles of baroclinic, barotropic and moist processes differ.

16. Line 283 “and range,” is this a typo? I am not sure what “range” means here.

R: “Range” is not a typo; it is the difference between the maximum and minimum values of each term (already defined in the caption of Table 1). We have now added a brief parenthetical note after the first occurrence to avoid confusion (L319-320).

17. Table 1 and Figure 3. What time are these values / pdfs valid at? e.g. averaged over the whole cyclone life cycle or at a given time e.g. time of maximum intensity? Please add this detail to the captions and the text at the start of section 3.1.

R: All statistics/PDFs are computed with every time-step of each cyclone, i.e. they represent averages over the whole life-cycle. This clarification has been added to the captions of Table 1 and Fig. 3 and to the opening sentence of Sect. 3.1 (L304-305).

18. Section 3.2 / Figure 4. The standard deviation values for all conversion terms are very large relative to the mean. This should be more clearly stressed in this section – it is only mentioned very briefly in lines 346-347.

R: Thank you for pointing this out. We now highlight the large spread of the conversion terms more clearly: several explicit references to the high standard deviation were added in Sect. 3.2 and a new opening sentence in Sect. 3.3 (L427-429) states that the EOF analysis is introduced precisely to disentangle this pronounced variability.

19. Figure 4. For a reader who is not very familiar with the Lorenz energy cycle diagrams, this figure is hard to understand as there are no labels indicating which box is which energy reservoir and what each conversion term is. The same comment applies to Figures 5- 8 and 12.

R: Agreed. All LEC diagrams (Figs. 4–8 & 12) have been redrawn with explicit labels for each energy reservoir and conversion pathway, following the layout already used in Fig. 2.

20. Line 383. Could the quantitative sum of the cumulative variance be added here to save a reader having to add it up? I think the first 8 EOFs explain 81.3% of the variance.

R: Added the cumulative variance of the first eight EOFs (81.3 %) to the text for immediate reference (L433).

21. Line 389. Could add that the first 4 EOFs explain 58.4% of the variance.

R: Inserted the cumulative variance explained by the first four EOFs (58.4 %) at the indicated location (L432).

22. Line 392 – 437. In this text where the EOFs are discussed it is hard to follow in places as it is unclear what the LEC terms are being compared to. Is it in comparison to Figure 4 and hence the full distribution of these terms? I also found this section hard to physically interpret as it is unclear to me if a cyclone only belongs to one EOF or if an individual cyclone can belong to different EOFs at different stages of its life cycle. Please can this be clarified.

R: All EOFs are computed from deviations with respect to the climatological-mean LEC shown in Fig. 4; hence, each EOF pattern represents a deviation (positive or negative) from that mean state. Every cyclone can be expressed as a linear combination of the EOFs, with its principal-component (PC) scores indicating how strongly each mode contributes to that cyclone's energetic structure. We have clarified in the revised version how the EOFs should be interpreted and how individual cyclones relate to them (L438-445).

23. Line 488 and 490 “three to seven fold increase...” what is this relative to? In line 488 “mean behavior” is stated but it is not clear if this is the mean of all cyclone or the mean of the intense cyclones. Please add clarifying details.

R: “Three- to seven-fold” is relative to the overall mean of all cyclones in the dataset. We now state this explicitly in the sentence (L556-558).

24. Figure 13. Please use the same colour bar between each panel to enable easier comparison.

R: Thank you for the suggestion. However, after the new grouping (see R25) cluster 4 contains far more tracks than the others; using a common scale would compress the signal in the remaining panels and mask their patterns. We therefore kept an individual colour-bar for each panel and note this choice in the caption for clarity. We have kept the same colour bar for Cluster 1 to 3, though.

25. Figure 14a. The total number of cyclones is 365 (adding up the numbers in this figure). However, in section 2.3, it was stated that only cyclones exceeding the 99th percentile of vorticity would be selected, so 1% of the 6789 cyclones which would be 68 cyclones. These numbers do not match up. Please can an explanation for this be given.

R: Thank you for pointing this out. After correcting a coding error in the track-selection routine, we chose to work with the 90-percentile threshold (instead of the 99 %) so the “intense-cyclone” sample is large enough for robust clustering. The section was rewritten and the new Figure 14 now shows 680 systems ($\approx 10\%$ of 6 789). All related numbers in the text (Section 3.5) and figures have been updated. We also updated the discussion of these results in Section 4.2 to reflect these changes (L699-714)

26. Section 4 in general is very long and repeats some of the material from section 3. Can this section be condensed?

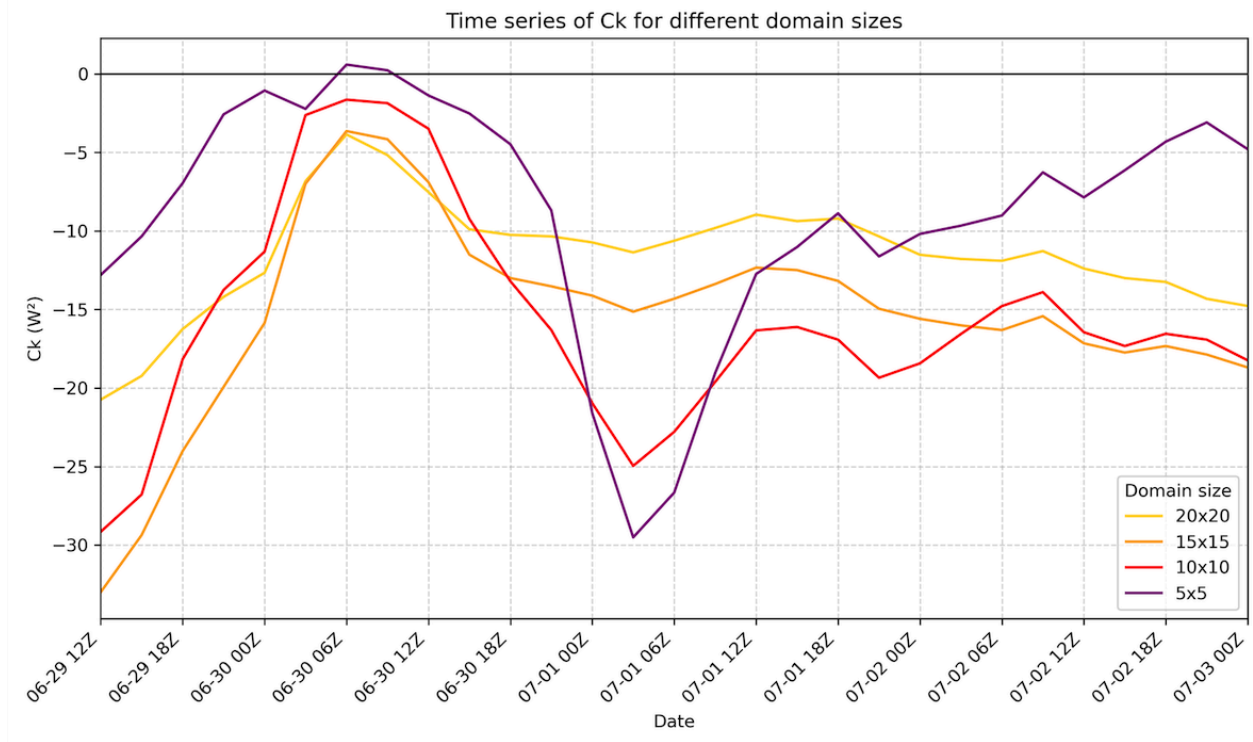
R: Section 4 was streamlined by deleting the repeated descriptions already covered in Section 3 and tightening the remaining text. Even after the new material requested in other comments was added, the revised Section 4 is now $\approx 20\%$ shorter.

27. One of the main results of this study is that the conversion from eddy kinetic energy to zonal kinetic energy is negative on average suggesting that kinetic energy is transferred from the zonal mean / the jets to the eddies. This contradicts global studies which show the transfer is in the opposite direction especially in the mature stage of cyclones, when the eddies feed back their energy on to the jet stream. This result is not explained well in this manuscript and I think it needs to be given how it differs considerably from previous studies. I wonder if this barotropic conversion term is partially caused by the cyclone data base also including some subtropical cyclones? Or is this an artifact of the semi-Lagrangian method?

R: The negative average CK reflects how we compute the decay phase, not an artefact of our data set or the semi-Lagrangian framework. In our climatology, CK reaches its most negative values during the mature stage, then becomes progressively less negative throughout decay; on average, it never turns positive (Fig. S4). Because our decay-phase mean is an average over several timesteps, the early, strongly negative values dominate and the phase-mean remains < 0 . In other words, during decay, CK continues to supply eddy kinetic energy that helps keep the cyclone “alive” until dissipation prevails and then the cyclone “dies”.

Previous case studies and Eulerian climatologies often (i) use broader, fixed domains and (ii) do not employ an objective life-cycle partition. Both choices can result in underestimations of barotropic conversions and blend mature and decay signals, which likely explains the sign difference you highlight. We now discuss this in Sect. 4.3 (L751-759) and point readers to Fig. S4 for the time-evolving behaviour. See below an example of how the domain size can

influence the CK term:



In the case above, we computed the LEC with the semi-Lagrangian framework using distinct domain sizes for the same System. Although this is an individual example, the same behaviour is consistent among distinct systems. As we highlighted in the Conclusion (L819-824), in future works we will explore why this happens - in fact, it is related to the second derivative of the vorticity term, which is not monotonic near the cyclone center and hence indicates barotropic instability occurrence (Rayleigh-Kuo criterion). These conclusions stem from the first author's doctoral thesis (listed in the references) and were not included in this paper, as it is already very extensive, but will be published in the following paper.

Sub-tropical cyclones are too few in our sample to bias the statistics (see last paragraph of Sect. 3.5); removing them does not change the sign of CK. Hence, the result primarily reflects the energetic evolution of extratropical cyclones in our semi-Lagrangian analysis.

Response to Reviewer 2

Overview:

This study uses a semi-lagrangian APE diagnostic to present a climatology of energy conversions for southwestern Atlantic cyclones. The authors present a characterization of the Lorenz energy cycle for cyclones in different regions and also make use of a principal component analysis to describe the major modes of variability among the cyclones. In my opinion, the methodology is well suited for the presented analysis and offers a valuable addition to the still very limited literature on the energetics of individual weather systems. However, the study would benefit from (1) clearer and more targeted visualizations of the key findings and (2) a better contextualization of your cyclone-centered APE approach and the results within the most recent literature. After those revisions, I'd be happy to recommend the manuscript for acceptance in Climate Dynamics.

Major comments:

1. Figures 4-8,12: The LEC figures are rather complex and feature a large amount of numerical values. I find it very hard to comprehend the in-text discussions in the figures and to compare the different EOFs with each other. Therefore, I suggest (1) to again

have the labels of the energy reservoirs and the conversion terms (as in Fig. 2; also see, e.g., Wang et al. (2024) their Fig. 8), (2) to somehow highlight the most important pathways that change and that you want to contrast for each EOF. Alternatively you could think about adding additional Figures, e.g. barplots, to directly compare selected energy pathways between the EOFs, such as the moist baroclinic chain you often mention, but I don't think this is strictly necessary.

R: This is an excellent suggestion. We have modified all the figures to ensure that the labels of the energy reservoirs are visible. For the EOF figures, we also changed the normalization of the arrow sizes to highlight the pathways with the highest energy flux, as we did in the other figures. Additionally, we have modified Figure 2 to emphasize the fluxes related to baroclinic conversions, barotropic conversions, and latent heat release to better guide the reader.

2. L702-708: I disagree with you that the presence of barotropic contributions to storm intensification are unexpected and I find the word 'unsettling' inappropriate. For instance, both Chemke et al. (2022) and Hadas & Kaspi (2025) report the importance of barotropic contributions to cyclone growth, which is especially relevant if one considers different storm locations relative to the upper-level shear (which is probably relevant for your regions as well). I suggest you better contextualize your results within the existing literature.

R: We have revised the section where we previously stated that the results were unexpected and "unsettling" and have reworded it to better reflect the findings (L716-723). Additionally, we have added a brief discussion on the studies by Chemke et al. (2022) and Hadas & Kaspi (2025), as recommended (L735-739). Thank you for the valuable literature suggestion.

Minor comments:

3. L74-84: Considering the limited amount of studies, I suggest you also discuss the most recent developments on local energetics. For instance, the local APE framework introduced by Novak & Tailleux (2018), which was applied in Liu et al. (2024) and Federer et al. (2024). In particular, Federer et al. (2025) presented a climatology of synoptic APE conversions for different life cycle stages of North Atlantic extratropical cyclones. They also highlight the relevance of APE advection into the storm tracks. How would this advection be reflected in your semi-lagrangian LEC? I assume this would be different for your regions, since they vary in latitude and with that the proximity to the Antarctic APE reservoir.

R: Thank you for suggesting these relevant references. We were not familiar with these studies, and they have certainly sparked our interest. We have added a discussion on this topic in the revised manuscript, particularly regarding the local APE framework and its potential application to our semi-lagrangian LEC (L760-771). We look forward to exploring this methodology more deeply in future studies.

4. L114-122: Did you use ERA5 on pressure or model levels? You mention the 137 model levels, but also indicate a pressure range.

R: We used ERA5 pressure levels in our analysis. The mention of 137 model levels was incorrect and has been corrected in the revised manuscript (L132-134).

5. L134: Have you also checked whether all the cyclones actually intensify (e.g., in terms of MSLP minimum or vorticity maximum)?

R: Yes, we verified that at least 95% of the cyclones exhibit an intensification phase (in terms of central maximum relative vorticity). This information has been added to the section describing the life cycle detection method in the revised manuscript (L251-254).

6. L150: Please be more specific regarding the data availability issue you encountered since this significantly impacted the relative distribution of your cyclones. Is this a limitation of ERA5 or of your tracking approach?

R: The data availability issue was related to the ERA5 download process, not to the reanalysis product itself or our tracking algorithm. During the second half of 2023 and the first half of 2024, ECMWF was migrating its data infrastructure and updating its API services. Throughout this period, the ERA5 API was frequently unstable, with slow data transfers and repeated interruptions caused by server outages or other unexplained errors. These technical problems often resulted in incomplete or failed downloads, despite multiple attempts and automated retries. As a result, some cyclones could not be included in the analysis. This limitation was therefore due to temporary external technical issues with the data access infrastructure, not with the underlying data or methods. We have clarified this in the revised manuscript (L163-174)

7. L177: APE diagnostics, and in particular their limitations, are non-trivial and the average reader is not familiar with the different frameworks out there. Therefore, I think it's important to outline at least the cornerstones of the framework used. In particular, I find the definition of the reference state, and a short discussion of the limitations introduced through the choice of local/global reference states important and would urge you to expand a little here.

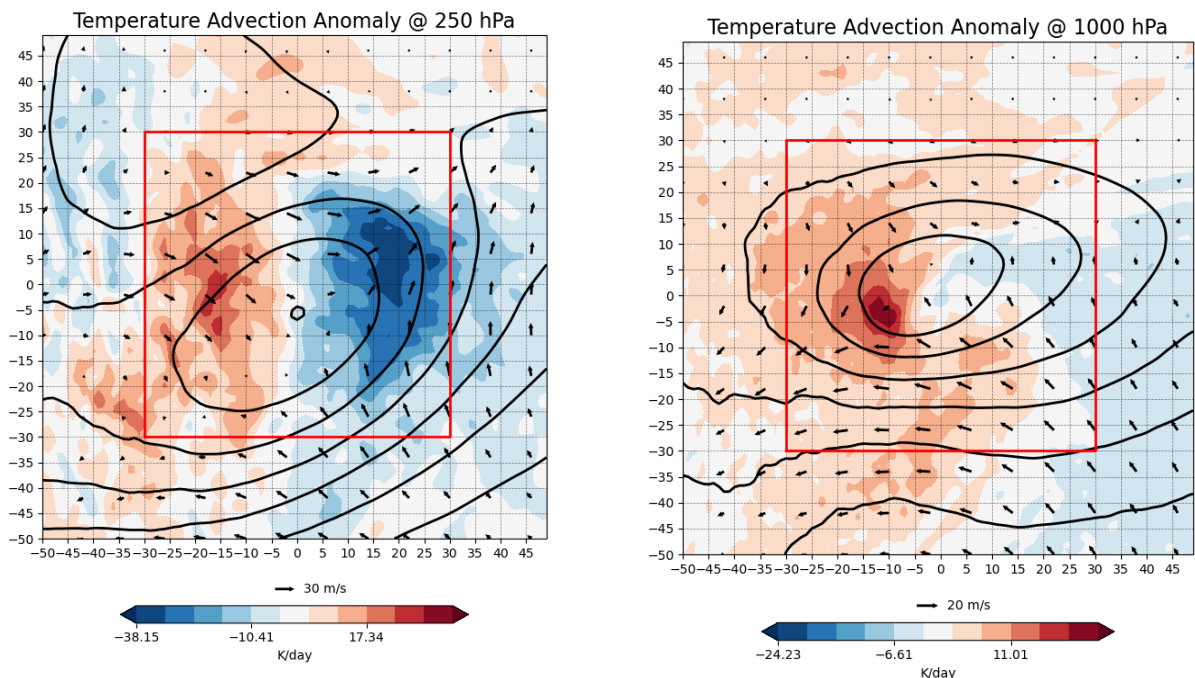
R: We added a discussion on the APE framework, including the reference state definition and its limitations, in the revised manuscript (L197-208).

8. L306/307: What would be your physical interpretation of the observed boundary flux terms? Same for the boundary pressure work. Could the boundary pressure work be interpreted as downstream flux of geopotential, as in downstream development? Also, why is the boundary pressure work so much larger than any conversion or boundary flux of energy? Does this boundary pressure work relate to the fact that your framework makes use of a local and a global reference state, and that there is an exchange of mass with your local reference state domain?

R: As mentioned at the end of the paragraph (349-350), the physical interpretation of the boundary pressure work terms is addressed in Section 4.1 (663-670). The interpretation of these terms is not straightforward and warrants further investigation in future studies.

We conducted preliminary tests to relate the boundary flux terms, such as the BAZ, to differential thermal advection between the eastern and western sides of the computational domain. This is illustrated in the figure below, which shows composites for the cyclones where the BAZ term was most intense, displaying the zonal deviation of temperature advection. It appears that at least BAZ is related to this feature. However, we concluded that a more in-depth analysis is needed to draw definitive conclusions on this. Given the length of the manuscript, we decided that this investigation was outside the scope of the current paper.

Regarding the boundary pressure work potentially relating to downstream development, we hypothesize that the BKe term could be more closely associated with this phenomenon. Again, further research is required to confirm this, and we plan to explore this topic in subsequent work.



9. L314-316: Diabatic heating in cyclones can significantly amplify their development. How would you interpret this statement in light of the near zero contribution of G_E ?

R: The contribution of G_E is not minimal. It is centered near zero (Figure 3), but it is positive on average (1.17 W/m^2) with a reasonable variability (± 3.14), indicating that there is variability from system to system in the importance of these terms, as mentioned in the passage in L351-355, as well as in L405-408. Additionally, there is also variability across the life cycle, as seen in Figure 4, mentioned in the passage in L394-397. This is discussed further in the paragraph of the discussion in L644-655.

Moreover, the contribution is more significant for the more intense cyclones, as can be seen in Figure 12, where intense cyclones always have a mean GE value higher than the mean of all cyclones, supporting your statement that diabatic contributions amplify cyclone development.

10. L348: General question regarding A_Z: How do you distinguish the import of A_Z into your region by the advection of the cyclone (active) vs. a poleward propagation of the cyclone itself (passive) toward higher mean values of A_Z?

We did not use a specific method to separate the import of AZ into the region by the active advection of the cyclone versus the passive poleward propagation of the cyclone itself toward higher mean values of AZ. Your question is also valid for KZ, which is likely influenced by the latitudinal displacement of the cyclones. However, we must consider the following: the current semi-Lagrangian framework does not explicitly account for the movement of the computational domain. Therefore, some of the observed temporal variability in the zonal terms (AZ and KZ) might be influenced by the displacement of the domain.

Nevertheless, cyclones generally tend to move poleward throughout their lifecycle, and we would expect a gradual and monotonic increase in AZ as the system moves into regions with higher background APE. For KZ, its maximum integrated value occurs near 60°S, meaning that cyclones originating in the ARG region and moving poleward would experience a monotonic decrease in KZ, while those forming in the SE-BR and LA-PLATA regions would likely experience a monotonic increase. Despite this general expectation, both AZ and KZ exhibit varying and non-monotonic behaviors during their lifecycle, as previously described. These variations suggest that the cyclones themselves strongly influence the local energetics. However, we cannot entirely rule out the effect of the cyclone's displacement on the behavior of these terms. We added L633-643 for considering this discussion.

11. L376: The results here are a bit hard to follow. What do you think about ending this section with a summary paragraph that highlights and contrasts again the most important variability among the EOFs, without discussing the details of each EOF separately? This would prepare the reader better for the following sections.

R: Suggestion accepted. A summary paragraph contrasting the most important variability among the EOFs without discussing the details of each EOF separately has been added (L492-505).

12. Figures 9,10: Since you discuss the origin regions, it would be helpful to again have at least the boxes (without labels) you showed in Fig. 1 for reference.

R: We have added the reference boxes (without labels) to Figures 9 and 10, as suggested.

13. Figure 15: Why is there no barotropic conversion from EKE to ZKE in the decay phase in your semi-lagrangian LEC?

R: While most previous studies report KE→KZ conversions during the decay phase of cyclones, this was not observed in our study. The main reason for this discrepancy is that in our results, CK is more negative during the mature phase, and instead of sharply transitioning to positive values during decay, it gradually becomes less negative, approaching neutrality by the end of the cyclone's life cycle. This suggests that during decay, CK plays a crucial role in sustaining the system before it completely dissipates. Additionally, the use of larger domains in Eulerian frameworks in previous studies may have underestimated the role of barotropic conversions, as seen in Black et al. (2013), where CK remains positive throughout the cyclone lifecycle. We added L751-759 to include this discussion

Technical corrections:

14. There's a typo in the title.

R: The typo in the title has been corrected. Thanks for noticing that.

15. L183: deception -> depiction

R: Typo corrected.

16. L185: Please check for consistent capitalization of "Semi-Lagrangian"

R: We adopted a consistent use of "Semi-Lagrangian"

17. L244: sentence unclear

R: We revised this sentence to make it clearer (L274-276).

References

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